

model-level operators
seeded as starting nodes
(objective, blend member)

budget phase machine:
exploit until every lineage plateaus
→ explore-burst: inject 5–8 fresh long-shot drafts
→ stop (hard cap 60 evaluated nodes; patience 20 evals after the burst without a new global best)

select parent lineage
(backtrack when a lineage plateaus:
3 children without improvement)

priors from
experience library

propose mutation:
solo moves: seed / hyperparameters / boundary push
blend moves: member set + weight search
(blends are first-class nodes over cached OOF predictions)

duplicate gate
(SHA-256 over the canonicalized config —
params, features, seed, blend members; a repeat burns budget, never re-evaluates)

data-level operators
each gets its own arm;
arms race, the winner
takes the full budget

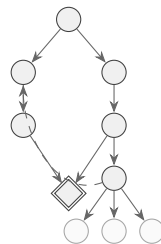
evaluate in sandboxed child process
(CV score; hang or crash → SIGKILL of
the process group, node marked failed,
search continues; blend cost guard)

add node, update global best
(a solo breakthrough reopens
plateaued blend lineages)

on stop:
best node →
submission

loop

example lineage



gray = plateaued lineage (3 misses)
dashed = backtrack after plateau
◊ = blend node (two parents)
double border = global best